08 Zerotrust

Lois-Kleinner Alpasan · Anticloud FZ LLE · PAX L5 Narrow L2 General 27B

Zero-Trust Local Mesh Benchmark — ALLENNLP

Company: Anticloud FZ LLE

1. Certificate Operations

Operation	Time
CA generation (4096-bit RSA)	1.8s (one-time)
Service cert issuance (2048-bit)	0.4s
mTLS handshake	2.1ms
Cert fingerprint (SHA-256)	0.1ms

2. Security Properties

Property	Value
CA key size	4096-bit RSA
Service cert key size	2048-bit RSA
Service cert TTL	1 day (configurable)
TLS version	1.3 (minimum)
Cipher suites	$\text{TLS}_{\text{AES_256_GCM_SHA384'}}$ $\text{TLS}_{\text{CHACHA20_POLY1305_SHA256}}$
mTLS enforcement	Both client and server certs required

3. Attack Resistance

Attack	Mitigation	Status
MITM on localhost	mTLS — attacker cannot present valid cert	MITIGATED
Stolen service cert	TTL 1 day — rotates automatically	MITIGATED

Attack	Mitigation	Status
CA key compromise	CA key never transmitted, air-gapped	MITIGATED
Downgrade to TLS 1.2	Minimum TLS 1.3 enforced	MITIGATED